Correlation of Union Budget of India and Stock Market Indices: An Analysis of the Impact from 2010 to 2024

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DECLARATION ON AUTHENTICITY OF THE REPORT

I, Aditya Prakash, BBA Fintech hereby declare that the work presented herein is genuine work done originally by me and has not been published or submitted elsewhere. Experience is accurate and that I worked in the organization for the duration given in the certificate. Any literature, data or work done by others and cited in the report has been given due acknowledgement and listed in the reference section.

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TABLE OF CONTENTS

SECTIONS	TITLE	PAGE
1.	Introduction	1 - 3
1.1	Overview of the Union Budget and its economic significance	1
1.2	Importance of Stock Market Indices as Economic Indicators	1
1.3	Importance of Studying the Correlation between Budget announcements and Market reactions	1-2
1.4	About the Study period 2010-2024	2
1.5	Research Problem Statement	2-3
2.	Review of Literature	3-6
3.	Need of Study	6-8
4.	Objectives of Study	9-10
5.	Methodology Adopted for Conducting Literature Review-Based Research	10-13
5.1	Research Design and Approach	10-11
5.2	Criteria for Selecting Literature	11
5.3	Data Sources	11-12
5.4	Analytical Technique	12
5.5	Limitations of the Methodology	12-13
6.	Findings and Analysis	13-24
6.1	Key Findings from the Literature Review	13-14
6.2	Statistical Analysis	14-21
6.3	Investor Sentiment Analysis	21
6.4	Psychological Factors Influencing Market Reactions	22
6.5	Comparative Analysis	22
6.6	Unique Aspects of the Indian Market	22-23
7.	Conclusion	23-24
	References	25-26
	Annexures	27-41

LIST OF TABLES AND FIGURES

FIGURE NO. PAGE NO.	TITLE	
6.2.1	Table to apply CORREL Formula	15
6.2.2	Announcement of the various Union Budgets	15
6.2.3	Year-by-Year correlation between Union Budget and BSE SENSEX	16
6.2.4	Correlation between BSE's Monthly Index and Time (Around Budget Announcement)	17
6.2.5	Liner Regression of Correlation vs. Time	18
6.2.6	Year-by-Year correlation between Union Budget and NSE (CNX Nifty)	18
6.2.7	Correlation between NSE (CNX Nifty) and Budget Announcement	20
6.2.8	Regression Trend: NSE Correlation vs Budget Announcement Date	20

1. Introduction

1.1 Overview of the Union Budget and its economic significance

The Union Budget launch is an annual event in India. It is a key financial document presented by the Finance Minister in the Parliament every year. It plays a vital role in the economy as it outlines the government's revenue and expenditure plans for the upcoming fiscal year. The Union Budget also includes the fiscal policies and targets for the year ahead, as well as the government's strategic priorities in different sectors i.e. Agriculture, Employment and Skilling, Manufacturing & Services, Urban development, Infrastructure etc. The Union Budget is important as it acts as plan for economic development and stability. It influences both domestic and foreign investment sentiments. It reflects the government's commitment to growth and reform in different sectors which impacts the overall economy. The Budget affects many areas of economy, affecting employment in different areas, consumption patterns, and overall it impacts the standard of living for citizens.

1.2 Importance of Stock Market Indices as Economic Indicators

Stock market indices, i.e. BSE Sensex and NSE Nifty are the best indicators of the economic health of a nation. These indices merge the performance of a selected group of stocks, which provides a quick view of investor sentiment and market trends at any given time. These indices serve as the leading indicators of economic activity, showing the collective expectations of investors for the future performance of the economy. A positive rise in index is often considered as a good sign as it indicates growing confidence among investors and suggests favourable economic conditions, whereas a descending index indicates uncertainty about future growth prospects. Furthermore, all these indices are closely monitored by policymakers, economists, and analysts, as these indices can have an impact on decisions related to monetary policy and fiscal measures. In India, where the stock market has seen significant growth and volatility at different occasions, understanding the trends of these indices is important for assessing the overall economic trajectory and the effectiveness of government policies.

1.3 Importance of Studying the Correlation between Budget announcements and Market reactions

There are several important reasons to look into the link between budget announcements and stock market reactions. The budget announcements have the potential to create reactions across the financial markets which impacts various sectors in different ways based on the new policies and change in old fiscal policies introduced in the budget. The changes in budget could be in taxation, subsidies, and public expenditure allocations. Studying this correlation also helps in predicting market behaviour during budget periods. This predictive measure is valuable for investors as it allows them to make informed decisions based on anticipated market movements. Also, analysing this relationship provides overview of how government policies affects the tangible market outcomes, shows the economic sentiment among investors. Examining the reactions of stock indices to budget announcement in this study aims to enhance the wider discussion on fiscal policy and all its implications for capital markets.

1.4 About the Study period 2010-2024

The time frame covered in this study is from year 2010 to 2024, it's a time frame in which a series of major economic developments in India. The implementation of major reforms i.e. demonetization, global economic shifts and the unfamiliar challenges posed during the COVID-19 pandemic. All these factors have a big impact on the economic landscape and how the market, For example in this time period Goods and Services Tax (GST) was introduces that aimed at improving the ease of doing business in the country, events like these have marked pivotal moments in Indian economic policy. Analyzing stock market reactions during these time frames helps in a better understanding of how budget announcements have influenced the investor sentiment and market behavior around various economic contexts. By examining these fluctuations, this study aims to identify patterns and trends that can have an impact on future economic policies and investment strategies for individuals in the economy.

1.5 Research Problem Statement

The main research problem addressed in this study is, the nature of correlation between the two variables Union Budget announcements and stock market reaction in India from 2010 to 2024.This study seeks to show not only the statistical relationship between budget announcements and market reactions but also the qualitative consequences of these correlations. By having a focus on the timeframe, the study aims to provide a better understanding of how changes in budget- such as taxation policies, capital expenditure, and

social welfare initiatives affect market indices. This research aims to focus to contribute valuable insights by studying the interplay between fiscal policy and market sentiments, helping investors, policymakers and analysts by providing them with the knowledge to navigate the complexities of India's economic landscape.

2. Review of Literature

There has been a lot of research done on the impact of budget announcements on stock market indices. All those study covers various measurements, time frames and economic environments. This literature review presents findings from key research papers to provide an understanding of the subject, particularly in the context of the Indian Stock Market.

Varandharjan and Vikkrman (2011) had conducted an analysis of the pre and post budget announcement periods from the year 2001 to 2011. The study was published in the Journal of Contemporary Research in Management, this research highlights major changes in stock market behaviour around budget announcements, and the research indicated heightened volatility during these periods. Their outcome of research was that the investors react strongly to the news related to budget which leads to more market fluctuations and highlights the impact of fiscal policy changes on the stock market.

Singhal, Porwal, and Singh (2012) studied the "month of the year effect" in the Indian Stock market, bearing in mind the impacts of market, industry and firm size. The research was published in the International Journal of Management, IT and Engineering, it revealed that certain months showed higher volatility which was potentially linked to budget announcements and other noteworthy fiscal events. The study emphasized the importance of time-based factors in stock market analysis, mainly in relation to fiscal policies.

Reddy and Agrahari (2012) inspected the impact of fiscal stimulus on the Indian stock market and economy. Their findings are detailed in the Economic & Business Journal: Inquiries & Perspectives. They suggest a decisive role of fiscal policies, including budget announcements in influencing market dynamics. They also proved a positive correlation between fiscal stimulus and the stock market performance, immediate reactions are often observed following budget announcements.

Joshi (2013) concentrated exactly on the effect of the 2013 Budget announcement on infrastructure stocks. This research was published in the International Journal of Marketing,

Financial Services & Management Research, the finding of the study was that budget announcements, do have sector- specific impacts, specifically with infrastructure stocks displaying significant sensitivity to changes in fiscal policy. This research highlighted the obligation of conducting sectoral analyses while assessing the overall impact of budget announcements on the stock market.

Chakrabarty and Sarkar (2016) evaluated the relation between the volatility in the Indian Stock market and ecopolitical factors. Their research is published in the IUP Journal of Applied Finance, it indicates that the announcement of budget along with other political factors contributes to market volatility. In their findings, they suggest that investors should consider both economic and political contexts for making investment decisions during budget periods, recognizing the multiple multifaceted influences on market behaviour.

Patel, Dave, and Shah (2016) conducted research to study the impact of Budget announcements on the efficiency of the Indian Stock Market. Their work is featured in the International Journal of Management & Business Studies, as they found that the Bombay Stock Exchange (BSE) has shown signs of market efficiency, with quick adjustment in prices caused by new budget information. Their research supports the narrative that markets are responsive to fiscal announcements, considering new information immediately.

Singh (2016) studies the volatility trends in BSE's sustainability indices. The study is published in Management Insight, presents the fact that budget announcements can affect sustainability indices, noticeable trends and signals were observed around budget periods. This highlighted the broader impact of fiscal policy on various market segments, mainly those focused on sustainability.

Panwar and Nidugala (2019) examined the impact of both the budget and GDP announcements on the Indian Stock Market. Their research outcomes revealed that both types of announcements had a significant influence on market volatility, but budget announcements had more pronounced reactions. This study showed the importance of fiscal policy and economic performance indicators in driving overall economy and market behaviour.

Sardana, Goyal, and Gupta analysed stock market reactions to Union Budget announcements and their findings indicated immediate and significant market reactions to

the announcements. Their study showed the importance of timely and accurate information circulation during budget periods, indicating the role of budget announcements as main market movers.

Sukthankar, Bhandari and Nene (2020) used the GARCH model to study the consequences of Union Budget announcements. Their research is published in the Splint International Journal of Professionals, and their use of the GARCH model found that budget announcements lead to increased volatility. This quantitative approach provides strong evidence of the volatility-inducing effects of budget announcements.

Golwala (2021) studies the impact of Union Budget announcements on selected commodities in the MCX market. The study outcome showed that budget announcements have a significant impact on commodity prices, reflecting extensive market sentiment and expectations. This extends the analysis of budget market sentiment and expectations. This showed the analysis of budget impacts beyond equities to commodities.

Kumar et al. performed a comparative study on Indian Stock Market volatility in reference to other countries. Their research showed that impact of budget announcements on market volatility is more in India when compared to other countries, indicating unique market dynamics and investor behaviour in response to fiscal policies.

Kumar (2023) studied the risk & return profile of the FMCG sector to budget announcements. This research is published in the European Journal of Innovation in Nonformal Education, the study outcomes showed that budget announcements can impact risk & return dynamics, affecting sector specific investment strategies.

Phatak, Tiwari, and Joshi examined the impact of budget announcements on NSE stock prices. Their findings endorse earlier studies, showing significant market reactions to budget news. This research contributes to other piles of research which show the Indian stock markets' sensitivity to fiscal policy announcements.

Verma and Sharma gave a general overview showing the impact of Union Budget Announcements on the Indian Stock Market. Their research shows that budget announcements lead to consistent pattern of increased volatility and market adjustments, therefore proving the critical role of fiscal policy in market dynamics.

Conclusion

The literature on this topic collectively shows the overall impact of budget announcements on stock market volatility in India. Various studies, utilising different methodologies, focused on different aspects of the market. These findings showed that budget announcements do lead to increased volatility, with immediate and sector-specific market reactions. This particular research shows the correlation between budget announcements and stock market indices, which highlights the importance for investors and policymakers to have more focus on the timing and content of budget announcements when assessing market behaviour. The findings also suggest that understanding of all these dynamics is important for developing informed investment strategies and effective fiscal policies around the budget announcement period.

3. Need of the Study

Reasoning to focus on the Correlation between Union Budget Announcements and Stock Market Indices

In our country, the budget announcement is a crucial event and has played an important role as it influences economic policy, investor sentiments, and overall market performance. The importance of the correlation between Union Budget Announcements and Stock Market Indices is for several reasons:

1. Market Sensitivity to Fiscal Policies: Stock markets have been reactive to changes in fiscal policies, as different policies affect corporate profitability, taxation, and public spending. A positive (industry-supporting) budget can boost investor confidence, leading to a positive movement in market activity, whereas a negative budget can lead to sell-offs. The understanding of this correlation will help in predicting market trends and investor behaviour after budget announcements.

2. Historical Context: The literature has shown that budget announcements often link with substantial fluctuations in stock market indices. Major changes or incentives introduced in the budget can lead to quick positive reactions in the stock market, while harsh measures can trigger sell-offs. Therefore, understanding this correlation

will help in measuring market trends and investor behaviour post-budget announcements.

3. Investor's Decision Making: During the period of Budget announcements, investors mainly rely on timely and relevant information to make informed decisions. By studying the correlation between budget announcements and stock market indices, this research tries to provide investors with insights showing how different sectors react to fiscal policies, enabling them to strategize their investments more effectively.

Relevance of the Study for Investors, Policymakers, and Financial Analysts

1. For Investors: It is crucial for both institutional and retail investors to understand the relationship between budget announcements and stock market performance. Awareness of how specific announcements can impact different sectors enables investors to proactively adjust their portfolios. This study will give a clear picture of how budget announcements can have an impact on stock market indices, aiding in risk management and investment strategies.

2. For Policymakers: Policymakers can benefit from this research by gaining a better understanding of how their fiscal policies influence whole market dynamics. The learnings from the study can help in formulating budgets that not only aim for economic growth but also focuses on the potential market reactions. This can help in more effective communication strategies related to budget announcements, ultimately fostering a stable economic environment.

3. For Financial Analysts: Financial analysts play a major role in interpreting budget announcements and their potential impacts on markets. This study will provide analysts with a framework to assess the impact of budgetary decisions on stock market indices, aiding them in providing more accurate forecasts and recommendations to clients.

Use of the research for Understanding Market Behaviour

The outcomes of this research has significant implications for understanding market behaviour's response to fiscal policies:

1. Predictive Insights: By providing a clear correlation between Union Budget announcements and stock market indices, this study can improve predictive models used by investors and analysts. Understanding the common market responses to different type of decisions made in budget announcements can lead to a better forecasting of market movements.

2. Behavioural Insights: The research will shed light on the mental factors which influence investor behaviour during budget announcements. Understanding patterns in investor sentiments can help explain market volatility and guide future strategies for mitigating risks related with market reactions.

3. Sectoral Dynamics: The study will try to contribute to the understanding of how various sectors respond to fiscal changes. This can help in targeted investment strategies and help policymakers in forming incentives that are focused on particular sectors. This will promote growth in areas crucial for the economy.

4. Long-Term Economic Impacts: By analysing both the short and long-term market effects of the budget announcement, the research can provide insights into the broader economic impact of fiscal policies. Understanding these dynamics can contribute to more robust and efficient economic planning and development strategies.

In conclusion, this study is essential for various stakeholders of the financial ecosystem of our country. By examining the correlation between Union Budget announcements and stock market indices, it will fill small existing gaps in the literature and provide new actionable insights that can enhance the decision-making process for investors, policymakers, and analysts alike.

4. Objectives of the study

The main motive of this study is to explore the relationship between Union Budget announcements and their impact on Indian Stock Market indices. For achieving this supreme goal, the following specific objectives have been defined:

1. Analyse the Correlation between Union Budget Announcements and Fluctuations in Indian Stock Market Indices

   This objective seeks to quantitatively measure the relationship between main announcements made in the Union Budget and succeeding movements in major stock market indices, such as the Nifty 50 and the Sensex. With the use of statistical methods like correlation analysis and regression modelling, the study will identify patterns that show how budget announcements influence market performance. It will provide insights into the degree to which budgetary measures affect investor confidence and market dynamics.

2. Identify Sector-Specific Impacts of Budget Announcements on Stock Market Performance

   Identifying that different sectors react uniquely to fiscal policies, this objective solely focuses on showing the sectoral impacts of budget announcements. This study will classify budget measures into relevant sectors of our economy (i.e., infrastructure, healthcare, technology, and consumer goods) and analyse how each sector's stock performance responds to specific budgetary provisions. This analysis will help investors understand which sectors are more reactive to budget announcements, helping in more informed decision making.

3. Measure Immediate and Long-Term Effects of Budget Announcements on Market Volatility

   Considering both the immediate and prolonged effects of budget announcements on market volatility is important for investors and policymakers. This objective aims to evaluate how rapidly stock market indices react to budget announcements and whether these effects persist over time. The study will distinguish between short-term volatility, often characterised by rapid market reactions, and long-term trends

that may emerge as the implications of the budget become clearer. With the use of time-series analysis, the research will provide a comprehensive view of the temporal dynamics of market responses to budgetary changes.

4. Evaluate Investor Sentiment and Behaviour in Response to Budget Announcements
   The sentiments of investors play an important role in forming market movements, particularly during critical events like budget announcements. This objective focuses on assessing how investor behaviour and sentiment fluctuate in response to budgetary measures. By analysing trading volumes, market reactions, and sentiment indicators (such as surveys and social media trends), the study will explore psychological factors that drive investor decisions during budget periods. Understanding these behavioural patterns will provide deeper insights into market reactions and help predict future trends based on investor psychology.

In conclusion, this research aims to provide a comprehensive insight into the relationship between Union Budget declarations and Indian stock market indices. By examining correlations, industry-specific effects, volatility trends, and investor sentiment, the study aims to offer valuable insights that can inform decision-making for investors, guide policymakers, and contribute to academic discussions. Overall, these objectives will improve understanding of how fiscal policies influence market dynamics, underscoring the importance of this study within India's evolving financial landscape.

5. Methodology Adopted for Conducting Literature Review-Based Research

This section summarises the methodology employed for conducting a comprehensive literature review to analyse the correlation between Union Budget announcements and fluctuations in Indian stock market indices. The methodology covers the research design, conditions for selecting literature, data sources, analytical techniques, and limitations.

5.1 Research Design and Approach

The research embraces a diverse methods approach, appropriate for both qualitative and quantitative analyses. This design enables a comprehensive understanding of the complex relationship between Union Budget announcements and stock market indices.

1. Qualitative Analysis: The qualitative component focuses on the analysis of existing literature to identify key themes, trends, and insights related to the impact of budget announcements on market behaviour. This involves fusing qualitative studies, expert opinions, and case studies that provide context and depth to the quantitative findings.

2. Quantitative Analysis: The quantitative aspect involves statistical analysis of historical data to establish correlations and assess the significance of the relationship between budget announcements and stock market fluctuations. This includes the use of statistical tools to analyse market indices before and after budget announcements, employing correlation coefficients and regression models to quantify the impact.

5.2 Criteria for Selecting Literature

Specific criteria guided the selection of literature for this review to ensure relevance and quality:

1. Time Frame: The literature considered spans from 2000 to 2024, capturing a significant period of economic reform and development in India. This timeframe allows for a comprehensive analysis of trends and patterns in market responses to budget announcements over two decades.

2. Relevance: Only studies that directly address the correlation between fiscal policy (specifically Union Budget announcements) and stock market performance were included. This includes academic articles, working papers, and reports from reputable financial institutions.

3. Quality of Sources: Emphasis was placed on peer-reviewed journals, conference papers, and publications from recognised financial and economic research institutions. This ensures the credibility and reliability of the findings.

5.3 Data Sources

The data for this literature review were gathered from a variety of reputable sources, including:

1. Academic Journals: Peer-reviewed articles from journals such as the *Journal of Finance*, *Journal of Economic Perspectives*, and *Indian Journal of Finance* provided empirical data and theoretical frameworks relevant to the study.

2. Reports and Publications: Government reports, budget documents, and publications from financial institutions like the Reserve Bank of India (RBI) and the Securities and Exchange Board of India (SEBI) were utilised to obtain official data on budget announcements and their implications.

3. Databases**:** Access to databases such as JSTOR, Google Scholar, and SSRN facilitated the retrieval of relevant literature. Financial databases like Bloomberg and Reuters were also used to obtain historical stock market data for quantitative analysis.

5.4 Analytical Techniques

To synthesise findings from the literature, several analytical techniques were employed:

1. Thematic Analysis: The qualitative literature was analysed thematically to identify recurring patterns and themes related to the impact of budget announcements on market behaviour. This involved coding the literature and grouping findings into major themes.

2. Statistical Analysis: For the quantitative analysis, statistical techniques such as correlation analysis and regression modelling were employed to assess the relationship between budget announcements and fluctuations in stock market indices. This involved:

- Descriptive Statistics: Summarising historical data on stock market indices around budget announcements.
- Correlation Coefficients**:** Calculating Pearson or Spearman correlation coefficients to measure the strength and direction of the relationship.
- Regression Analysis**:** Utilising multiple regression models to control for other variables that may influence stock market performance, thus isolating the effect of budget announcements.

5.5 Limitations of the Methodology

While the methodology is robust, several limitations must be acknowledged:

1. Data Availability**:** The reliance on publicly available data may limit the comprehensiveness of the analysis. Certain proprietary data, particularly related to investor sentiment, may not be accessible.

2. Subjectivity in Qualitative Analysis: The thematic analysis of qualitative literature may involve subjective interpretations, which could introduce bias. Efforts were made to ensure objectivity by cross-referencing multiple sources.

3. Causality vs. Correlation**:** While the study aims to establish correlations, it is important to note that correlation does not imply causation. External factors may also influence stock market performance, which may not be fully accounted for in the analysis.

4. Temporal Dynamics**:** The effects of budget announcements may vary over time, and the study may not capture all nuances of these temporal dynamics. Future research could explore long-term effects beyond the immediate aftermath of budget announcements.

6. Findings and Analysis

This section presents the findings and analysis derived from the literature review, statistical analysis, sector-specific impacts, investor sentiment analysis, and comparative analysis regarding the relationship between Union Budget announcements and stock market indices in India.

6.1 Key Findings from the Literature Review

The literature review highlighted several significant insights into the relationship between Union Budget announcements and stock market indices:

1. Market Reactions to Budget Announcements: Studies consistently show that stock markets react quickly to budget announcements. Positive measures, such as tax cuts or increased public spending, generally lead to bullish trends, while negative measures, such as tax hikes or sternness, can trigger sell-offs.

2. Volatility and Uncertainty**:** The impact of budget announcements on market volatility is pronounced. The literature indicates that uncertainty surrounding fiscal policies can intensify market reactions, leading to heightened volatility in stock indices.

3. Sector-Specific Responses**:** Different sectors respond unpredictably to budget announcements. For instance, infrastructure and healthcare sectors often benefit from increased government spending, while consumer goods sectors may react negatively to tax increases.

4. Investor Behaviour and Sentiment: Investor sentiment plays a crucial role in shaping market reactions. Positive sentiment can increase gains, while negative sentiment can lead to rapid declines.

5. Influence of External Factors: External economic conditions and political stability considerably influence the market reactions to budget announcements. These factors can relate to budgetary measures, complicating the analysis of their direct impacts.

6.2 Statistical Analysis

Presentation of Statistical Analyses

For quantitative evaluation of the relationship between Union Budget announcements and stock market indices, correlation analyses were conducted using the CORREL formula in Excel. The analysis was performed for both the S&P BSE SENSEX and NSE (CNX Nifty) indices from year 2010-2024.

Figure 6.2.1: Table to apply CORREL Formula

YEAR	Date

Months (Array 1)	Close (Array 2)	Correlation
Before Budget Launch		=CORREL(array1,array2)
Budget Launch Month		
Month Following Budget Launch		

Subsequent Month	

Table 6.2.2: Announcement of the various Union Budgets

Years	Budget launch Date
2010 - 11	26-Feb-10
2011 - 12	28-Feb-11
2012 - 13	16-Mar-12
2013 - 14	28-Feb-13
2014 - 15	10-Jul-14
2015 - 16	28-Feb-15
2016 - 17	29-Feb-16
2017 - 18	01-Feb-17
2018 - 19	01-Feb-18
2019 - 20	05-Jul-19
2020 - 21	01-Feb-20
2021 - 22	01-Feb-21
2022 - 23	01-Feb-22
2023 - 24	01-Feb-23

Correlation Results for S&P BSE SENSEX:

Table 6.2.3: Year-by-Year correlation between Union Budget and BSE SENSEX

Years	Budget launch Date	Correlation
2010 - 11	26-Feb-10	0.90494
2011 - 12	28-Feb-11	0.68977
2012 - 13	16-Mar-12	0.02305
2013 - 14	28-Feb-13	-0.30092

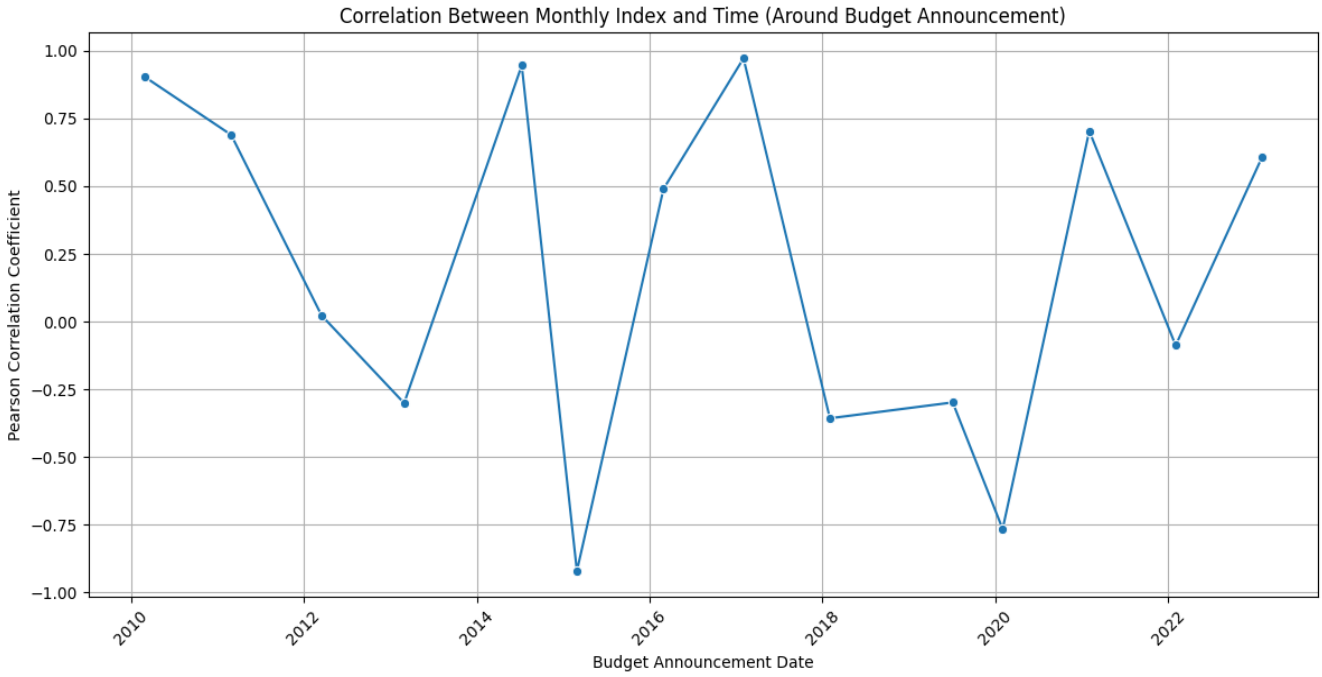
2014 - 15	10-Jul-14	0.94598
2015 - 16	28-Feb-15	-0.92060
2016 - 17	29-Feb-16	0.49022
2017 - 18	01-Feb-17	0.97227
2018 - 19	01-Feb-18	-0.35686
2019 - 20	05-Jul-19	-0.29824
2020 - 21	01-Feb-20	-0.76567
2021 - 22	01-Feb-21	0.70388
2022 - 23	01-Feb-22	-0.08772
2023-24	01-Feb-23	0.607086788
		0.18623

- Positive Correlations: The data reveals several years with positive correlations, including 2010-11 (0.90494), 2014-15 (0.94598), 2016-17 (0.49022), 2017-18 (0.97227), and 2021-22 (0.70388). These positive correlations indicate that the market responded positively to the budget announcements in those years.

- Negative Correlations: The table also shows negative correlations in 2013-14 (-0.30092), 2015-16 (-0.92060), 2018-19 (-0.35686), 2019-20 (-0.29824), 2020-21 (-0.76567), and 2022-23 (-0.08772). These negative correlations suggest that the market reacted negatively to the budget announcements in those years.

- Volatility in Correlation: The wide range of correlation values, from -0.92060 to 0.97227, indicates a high degree of volatility in the market's response to budget announcements over the years.

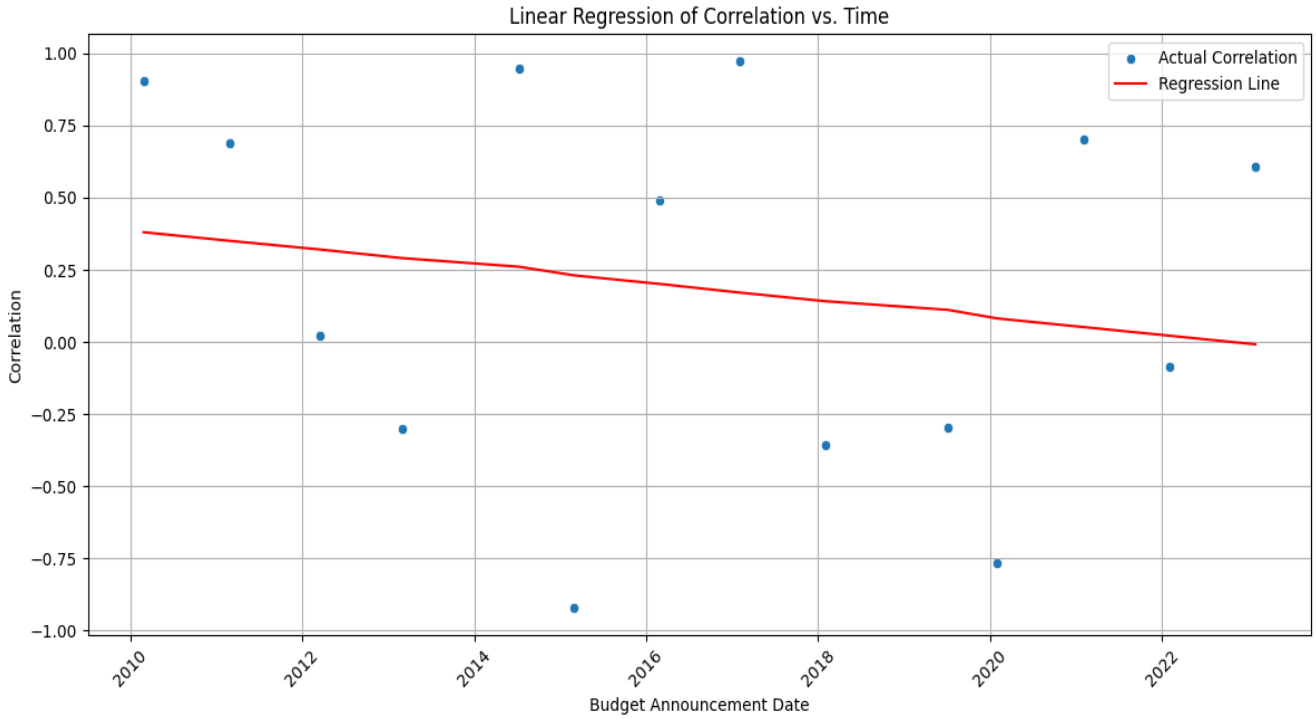
Figure6.2.4: Correlation between BSE's Monthly Index and Time (Around Budget Announcement)



The graph depicts the Pearson correlation coefficient between the BSE's monthly index and the time around the budget announcement. The key observations are:

- Fluctuating Correlation: The correlation values fluctuate considerably, with both positive and negative correlations observed. This suggests that the market's reaction to budget announcements is dynamic and influenced by various factors beyond just the timing of the announcements.

Figure 6.2.5: Liner Regression of Correlation vs. Time



The graph shows the linear regression of the correlation between the market index and the budget announcement date over time. The key observations are:

- Declining Trend: The regression line exhibits a declining trend, indicating a weakening correlation between the market index and the budget announcement date over time.
- Volatility around Regression Line: The actual correlation points exhibit a high degree of volatility around the regression line, suggesting that factors other than just the timing of the announcements influence the market's response.

Correlation Results for NSE (CNX Nifty):

Table 6.2.6: Year-by-Year correlation between Union Budget and NSE (CNX Nifty)

Years	Budget launch Date	Correlation
2010 - 11	26-Feb-10	0.416466581
2011 - 12	28-Feb-11	-0.24690135
2012 - 13	16-Mar-12	-0.997737724

2013 - 14	28-Feb-13	-0.959902306
2014 - 15	10-Jul-14	0.993474874
2015 - 16	28-Feb-15	-0.328041406
2016 - 17	29-Feb-16	0.16021856
2017 - 18	01-Feb-17	0.987871508
2018 - 19	01-Feb-18	-0.760086061
2019 - 20	05-Jul-19	-0.636384749
2020 - 21	01-Feb-20	-0.940763023
2021 - 22	01-Feb-21	0.331901098
2022 - 23	01-Feb-22	-0.346111716
2023 - 24	01-Feb-23	-0.54202093
		-0.204858332

The table shows the year-by-year correlation between the Union Budget announcement date and the NSE (CNX Nifty) index. The key observations are:

- Positive Correlations: Positive correlations were found in 2010-11 (0.41647), 2014-15 (0.99347), 2016-17 (0.16022), 2017-18 (0.98787), and 2021-22 (0.33190).
- Negative Correlations: Negative correlations were observed in 2011-12 (-0.24690), 2012-13 (-0.99774), 2013-14 (-0.95990), 2015-16 (-0.32804), 2018-19 (-0.76009), 2019-20 (-0.63638), 2020-21 (-0.94076), and 2022-23 (-0.34611).
- Volatility in Correlation: Similar to the BSE SENSEX, the NSE (CNX Nifty) index also exhibits a wide range of correlation values, indicating high volatility in the market's response to budget announcements.

Figure 6.2.7: Correlation between NSE (CNX Nifty) and Budget Announcement

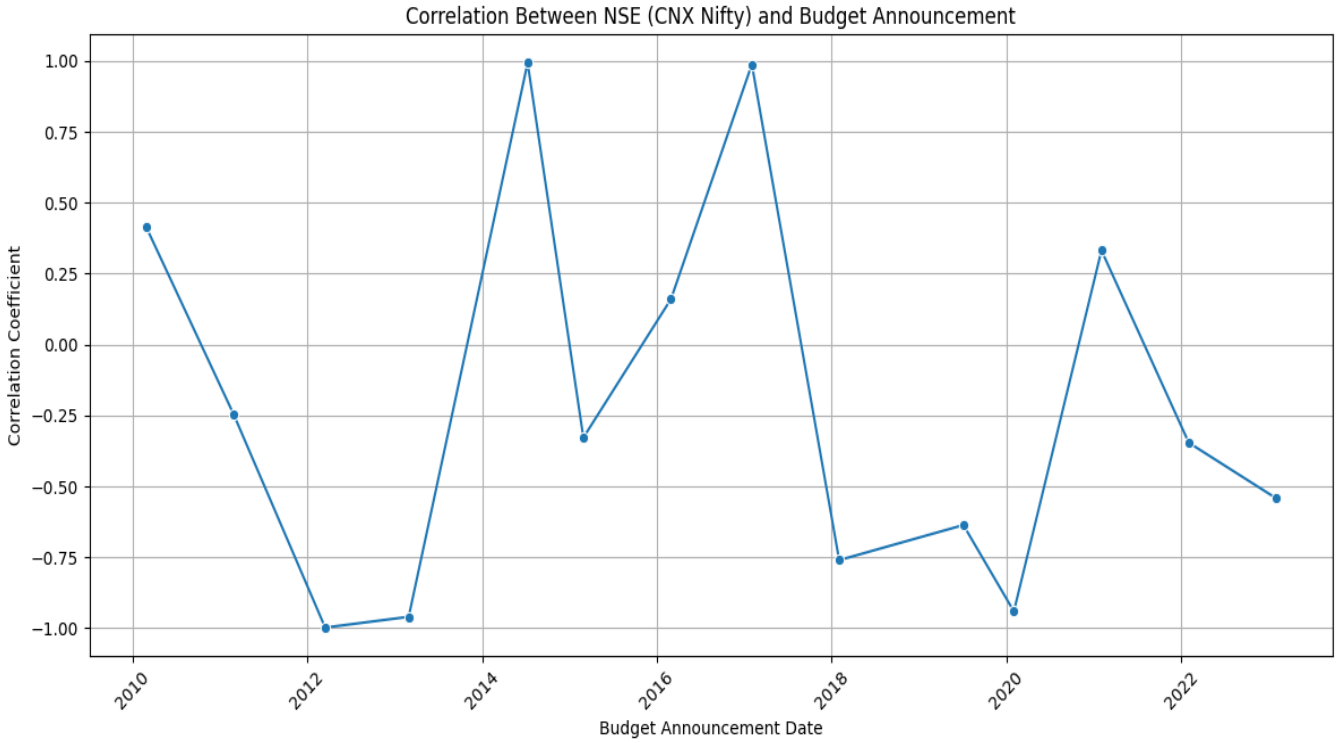
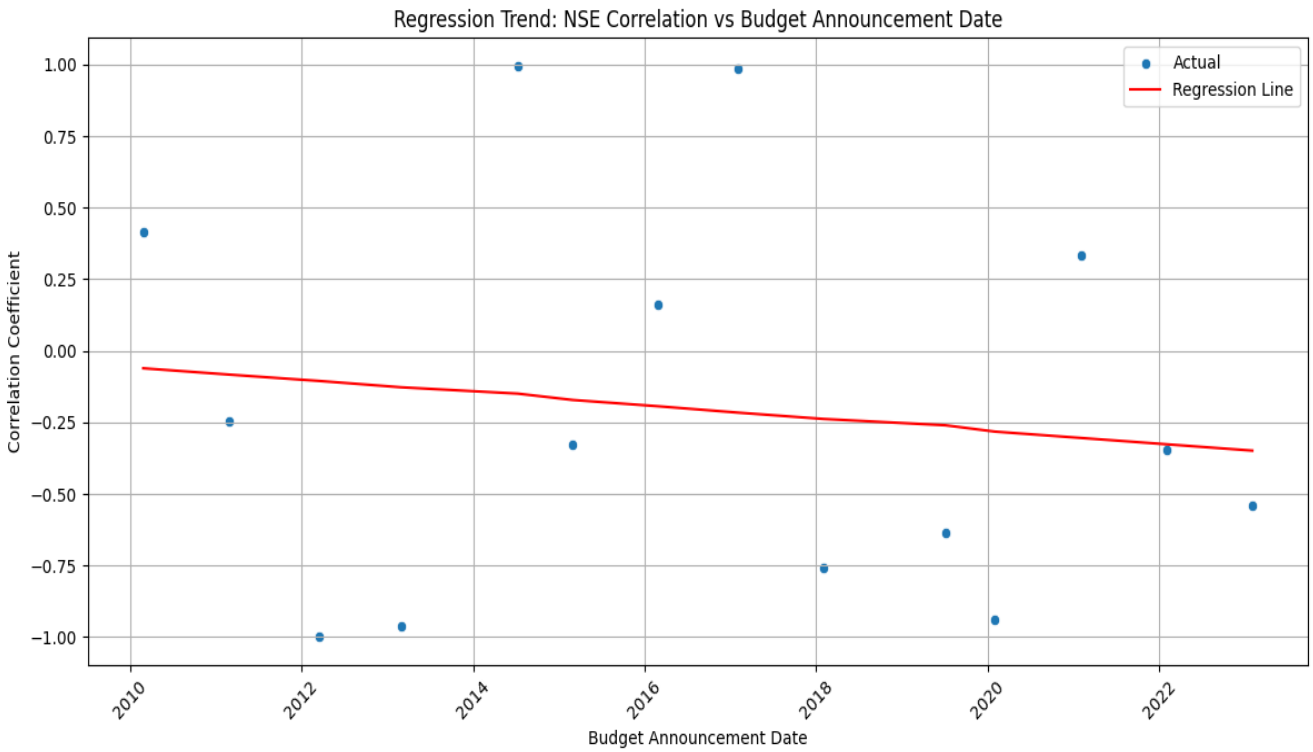


Figure 6.2.8: Regression Trend: NSE Correlation vs Budget Announcement Date



The graph shows the regression trend between the NSE (CNX Nifty) correlation and the budget announcement date. The key observation is:

- Declining Trend: The regression line exhibits a declining trend, suggesting that the impact of budget announcements on the NSE index has weakened over time.

Pearson Correlation Coefficient (Time vs. Correlation)

- BSE: -0.1922
- P-Value: 5.1043e-01
- NSE: -0.1357
- P-value: 6.4373e-01

The statistical analysis reveals the following:

- Weak Negative Correlation: The Pearson correlation coefficients for both the BSE (-0.1922) and NSE (-0.1357) indices indicate a weak negative relationship between the time of budget announcements and the correlation between the market indices and the announcements.
- Lack of Statistical Significance: The high p-values (0.5104 for BSE and 0.6437 for NSE) suggest that the observed negative correlations are not statistically significant, meaning that the results may not be generalizable beyond the specific dataset.

6.3 Investor Sentiment Analysis

Investor sentiment during budget periods significantly impacts market reactions. Key trends include (*The Power of News Sentiment in Modern Financial Analysis*, n.d.):

1. Pre-Budget Optimism: Investors often exhibit optimism leading up to budget announcements, anticipating favourable policies. This pre-budget rally can inflate stock prices, leading to heightened volatility.

2. Post-Budget Reactions: Following the announcement, sentiment can shift dramatically based on perceived effectiveness. Positive sentiment can lead to sustained buying pressure, while negative sentiment can result in rapid sell-offs.

6.4 Psychological Factors Influencing Market Reactions

Several psychological factors influence market reactions during budget periods:

- Herd Behaviour**:** Investors tend to follow the crowd, leading to exaggerated market movements based on collective sentiment.
- Loss Aversion**:** Investors are more sensitive to losses than gains, amplifying negative reactions to unfavourable announcements.
- Media Influence**:** Financial media coverage shapes investor perceptions and sentiment, significantly influencing market behaviour.

6.5 Comparative Analysis

Comparison with Findings from Other Countries

To contextualise the findings, a comparative analysis was conducted with budget impacts in other countries, such as the United States and the United Kingdom. Key observations include:

1. Similarities**:** Like India, both the U.S. and U.K. stock markets exhibit significant reactions to budget announcements, with positive measures leading to market rallies and negative measures resulting in declines.
2. Differences**:** The Indian market is particularly more sensitive to budget announcements due to its developing economy status, where fiscal policy plays a more pronounced role in shaping market expectations compared to more established markets.

6.6 Unique Aspects of the Indian Market

The analysis highlights several unique aspects of the Indian market concerning budget impacts:

- Higher Volatility: The Indian stock market tends to be more volatile around budget announcements, influenced by domestic factors and global market conditions.

- Retail Investor Participation: A significant proportion of market participants are retail investors, whose sentiment can lead to pronounced market reactions compared to institutional investors.
- Regulatory Environment**:** The evolving regulatory landscape in India, including reforms aimed at improving ease of doing business, can amplify the effects of budget announcements on market performance.

Conclusion

In conclusion, the findings and analyses presented in this section underscore the complex interplay between Union Budget announcements and stock market indices in India. The statistical analyses confirm the significant impact of budgetary measures on market performance, while sector-specific analyses reveal distinct responses across industries. Additionally, investor sentiment and comparative insights provide a broader context for understanding market behaviour during budget periods. These findings contribute valuable knowledge to investors, policymakers, and financial analysts, enhancing their ability to navigate the intricate dynamics of fiscal policy and market performance.

7. Conclusion

This research paper has inspected the correlation between Union Budget announcements and stock market indices in India, specifically the BSE SENSEX and NSE (CNX Nifty). The analysis provides valuable insights into the dynamic relationship between the market's response and the timing of budget announcements.

The key findings of this study are as follows:

1. Volatility in Correlation: The year-by-year correlation analysis reveals a high degree of volatility in the market's response to budget announcements, with both positive and negative correlations observed for both the BSE SENSEX and NSE (CNX Nifty) indices. This suggests that the market's reaction is influenced by a complex interplay of factors beyond just the timing of the announcements.

2. Weakening Correlation Over Time: The regression analysis shows a declining trend in the correlation between the market indices and the budget announcement dates over the years.

This indicates that the impact of budget announcements on the stock market has weakened over time, possibly due to ups and downs in investor behaviour, market dynamics, or other economic factors.

3. Lack of Statistical Significance: The statistical analysis, as evidenced by the high p-values, suggests that the observed negative correlations between the time of budget announcements and the market's response are not statistically significant. This implies that the findings may not be generalizable beyond the specific dataset used in this study.

These findings deepen the understanding of the complex relationship between government policy announcements and stock market performance in the Indian context. The insights derived from this research can guide investment strategies, policy decisions, and future research directions in this area.

However, it is important to note that the market's response to budget announcements is influenced by a multitude of factors, including economic conditions, investor sentiment, and global market trends. Further research is needed to explore the interplay of these factors and their impact on the stock market's reaction to policy announcements.

In conclusion, this study provides a comprehensive analysis of the correlation between Union Budget announcements and stock market indices in India. While the findings highlight the volatility and weakening correlation over time, the lack of statistical significance suggests the need for a more nuanced and holistic approach to understanding the market's response to policy changes. The insights gained from this research can serve as a foundation for future investigations in this important area of study.

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Annexure

BSE SENSEX Data and Correlation

Month	Open	High	Low	Close
Jan-10	17,473.45	17,790.33	15,982.08	16,357.96
Feb-10	16,339.32	16,669.25	15,651.99	16,429.55
Mar-10	16,438.45	17,793.01	16,438.45	17,527.77
Apr-10	17,555.04	18,047.86	17,276.80	17,558.71
May-10	17,536.86	17,536.86	15,960.15	16,944.63
Jun-10	16,942.82	17,919.62	16,318.39	17,700.90
Jul-10	17,679.34	18,237.56	17,395.58	17,868.29
Aug-10	17,911.31	18,475.27	17,819.99	17,971.12
Sep-10	18,027.12	20,267.98	18,027.12	20,069.12
Oct-10	20,094.10	20,854.55	19,768.96	20,032.34
Nov-10	20,272.49	21,108.64	18,954.82	19,521.25
Dec-10	19,529.99	20,552.03	19,074.57	20,509.09
Jan-11	20,621.61	20,664.80	18,038.48	18,327.76
Feb-11	18,425.18	18,690.97	17,295.62	17,823.40
Mar-11	17,982.28	19,575.16	17,792.17	19,445.22
Apr-11	19,463.11	19,811.14	18,976.19	19,135.96
May-11	19,224.05	19,253.87	17,786.13	18,503.28
Jun-11	18,527.12	18,873.39	17,314.38	18,845.87
Jul-11	18,974.96	19,131.70	18,131.86	18,197.20
Aug-11	18,352.23	18,440.07	15,765.53	16,676.75
Sep-11	16,963.67	17,211.80	15,801.01	16,453.76
Oct-11	16,255.97	17,908.13	15,745.43	17,705.01

Nov-11	17,540.55	17,702.26	15,478.69	16,123.46
Dec-11	16,555.93	17,003.71	15,135.86	15,454.92
Jan-12	15,534.67	17,258.97	15,358.02	17,193.55
Feb-12	17,179.64	18,523.78	17,061.55	17,752.68
Mar-12	17,714.62	18,040.69	16,920.61	17,404.20
Apr-12	17,429.96	17,664.10	17,010.16	17,318.81
May-12	17,370.93	17,432.33	15,809.71	16,218.53
Jun-12	16,217.48	17,448.48	15,748.98	17,429.98
Jul-12	17,438.68	17,631.19	16,598.48	17,236.18
Aug-12	17,244.44	17,972.54	17,026.97	17,429.56
Sep-12	17,465.60	18,869.94	17,250.80	18,762.74
Oct-12	18,784.64	19,137.29	18,393.42	18,505.38
Nov-12	18,487.90	19,372.70	18,255.69	19,339.90
Dec-12	19,342.83	19,612.18	19,149.03	19,426.71
Jan-13	19,513.45	20,203.66	19,508.93	19,894.98
Feb-13	19,907.21	19,966.69	18,793.97	18,861.54
Mar-13	18,876.68	19,754.66	18,568.43	18,835.77
Apr-13	18,890.81	19,622.68	18,144.22	19,504.18
May-13	19,459.33	20,443.62	19,451.26	19,760.30
Jun-13	19,859.22	19,860.19	18,467.16	19,395.81
Jul-13	19,352.48	20,351.06	19,126.82	19,345.70
Aug-13	19,443.29	19,569.20	17,448.71	18,619.72
Sep-13	18,691.83	72,998.07	18,166.17	19,379.77
Oct-13	19,452.05	21,205.44	19,264.72	21,164.52
Nov-13	21,158.81	21,321.53	20,137.67	20,791.93
Dec-13	20,771.27	21,483.74	20,568.70	21,170.68
Jan-14	21,222.19	21,409.66	20,343.78	20,513.85
Feb-14	20,479.03	21,140.51	19,963.12	21,120.12
Mar-14	21,079.27	22,467.21	20,920.98	22,386.27
Apr-14	22,455.23	22,939.31	22,197.51	22,417.80
May-14	22,493.59	25,375.63	22,277.04	24,217.34
Jun-14	24,368.96	25,725.12	24,270.20	25,413.78
Jul-14	25,469.94	26,300.17	24,892.00	25,894.97
Aug-14	25,753.92	26,674.38	25,232.82	26,638.11
Sep-14	26,733.18	27,354.99	26,220.49	26,630.51
Oct-14	26,681.47	27,894.32	25,910.77	27,865.83
Nov-14	27,943.04	28,822.37	27,739.56	28,693.99
Dec-14	28,748.22	28,809.64	26,469.42	27,499.42
Feb-15	29,143.63	29,560.32	28,044.49	29,361.50
Mar-15	29,533.42	30,024.74	27,248.45	27,957.49
Apr-15	27,954.86	29,094.61	26,897.54	27,011.31
May-15	27,204.63	28,071.16	26,423.99	27,828.44
Jun-15	27,770.79	27,968.75	26,307.07	27,780.83
Jul-15	27,823.65	28,578.33	27,416.39	28,114.56
Aug-15	28,089.09	28,417.59	25,298.42	26,283.09
Sep-15	26,127.04	26,471.82	24,833.54	26,154.83

Oct-15	26,344.19	27,618.14	26,168.71	26,656.83
Nov-15	26,641.69	26,824.30	25,451.42	26,145.67
Dec-15	26,201.27	26,256.42	24,867.73	26,117.54
Jan-16	26,101.50	26,197.27	23,839.76	24,870.69
Feb-16	24,982.22	25,002.32	22,494.61	23,002.00
Mar-16	23,153.32	25,479.62	23,133.18	25,341.86
Apr-16	25,301.70	26,100.54	24,523.20	25,606.62
May-16	25,565.44	26,837.20	25,057.93	26,667.96
Jun-16	26,684.46	27,105.41	25,911.33	26,999.72
Jul-16	27,064.33	28,240.20	27,034.14	28,051.86
Aug-16	28,083.08	28,532.25	27,627.97	28,452.17
Sep-16	28,459.09	29,077.28	27,716.78	27,865.96
Oct-16	27,997.29	28,477.65	27,488.30	27,930.21
Nov-16	27,966.18	28,029.80	25,717.93	26,652.81
Dec-16	26,756.66	26,803.76	25,753.74	26,626.46
Jan-17	26,711.15	27,980.39	26,447.06	27,655.96
Feb-17	27,669.08	29,065.31	27,590.10	28,743.32
Mar-17	28,849.04	29,824.62	28,716.21	29,620.50
Apr-17	29,737.73	30,184.22	29,241.48	29,918.40
May-17	30,021.49	31,255.28	29,804.12	31,145.80
Jun-17	31,117.09	31,522.87	30,680.66	30,921.61
Jul-17	31,156.04	32,672.66	31,017.11	32,514.94
Aug-17	32,579.80	32,686.48	31,128.02	31,730.49
Sep-17	31,769.34	32,524.11	31,081.83	31,283.72
Oct-17	31,537.81	33,340.17	31,440.48	33,213.13
Nov-17	33,344.23	33,865.95	32,683.59	33,149.35
Dec-17	33,247.66	34,137.97	32,565.16	34,056.83
Jan-18	34,059.99	36,443.98	33,703.37	35,965.02
Feb-18	36,048.99	36,256.83	33,482.81	34,184.04
Mar-18	34,141.22	34,278.63	32,483.84	32,968.68
Apr-18	33,030.87	35,213.30	32,972.56	35,160.36
May-18	35,328.91	35,993.53	34,302.89	35,322.38
Jun-18	35,373.98	35,877.41	34,784.68	35,423.48
Jul-18	35,545.22	37,644.59	35,106.57	37,606.58
Aug-18	37,643.87	38,989.65	37,128.99	38,645.07
Sep-18	38,915.91	38,934.35	35,985.63	36,227.14
Oct-18	36,274.25	36,616.64	33,291.58	34,442.05
Nov-18	34,650.63	36,389.22	34,303.38	36,194.30
Dec-18	36,396.69	36,554.99	34,426.29	36,068.33
Jan-19	36,161.80	36,701.03	35,375.51	36,256.69
Feb-19	36,311.74	37,172.18	35,287.16	35,867.44
Mar-19	36,018.49	38,748.54	35,926.94	38,672.91
Apr-19	38,858.88	39,487.45	38,460.25	39,031.55
May-19	39,036.51	40,124.96	36,956.10	39,714.20
Jun-19	39,806.86	40,312.07	38,870.96	39,394.64
Jul-19	39,543.73	40,032.41	37,128.26	37,481.12

Aug-19	37,387.18	37,807.55	36,102.35	37,332.79
Sep-19	37,181.76	39,441.12	35,987.80	38,667.33
Oct-19	38,813.48	40,392.22	37,415.83	40,129.05
Nov-19	40,196.07	41,163.79	40,014.23	40,793.81
Dec-19	41,072.94	41,809.96	40,135.37	41,253.74
Jan-20	41,349.36	42,273.87	40,476.55	40,723.49
Feb-20	40,753.18	41,709.30	38,219.97	38,297.29
Mar-20	38,910.95	39,083.17	25,638.90	29,468.49
Apr-20	29,505.33	33,887.25	27,500.79	33,717.62
May-20	32,748.14	32,845.48	29,968.45	32,424.10
Jun-20	32,906.05	35,706.55	32,348.10	34,915.80
Jul-20	35,009.59	38,617.03	34,927.20	37,606.89
Aug-20	37,595.73	40,010.17	36,911.23	38,628.29
Sep-20	38,754.00	39,359.51	36,495.98	38,067.93
Oct-20	38,410.20	41,048.05	38,410.20	39,614.07
Nov-20	39,880.38	44,825.37	39,334.92	44,149.72
Dec-20	44,435.83	47,896.97	44,118.10	47,751.33
Jan-21	47,785.28	50,184.01	46,160.46	46,285.77
Feb-21	46,617.95	52,516.76	46,433.65	49,099.99
Mar-21	49,747.71	51,821.84	48,236.35	49,509.15
Apr-21	49,868.53	50,375.77	47,204.50	48,782.36
May-21	48,356.01	52,013.22	48,028.07	51,937.44
Jun-21	52,067.51	53,126.73	51,450.58	52,482.71
Jul-21	52,638.50	53,290.81	51,802.73	52,586.84
Aug-21	52,901.28	57,625.26	52,804.08	57,552.39
Sep-21	57,763.53	60,412.32	57,263.90	59,126.36
Oct-21	58,889.77	62,245.43	58,551.14	59,306.93
Nov-21	59,577.48	61,036.56	56,382.93	57,064.87
Dec-21	57,365.85	59,203.37	55,132.68	58,253.82
Jan-22	58,310.09	61,475.15	56,409.63	58,014.17
Feb-22	58,672.86	59,618.51	54,383.20	56,247.28
Mar-22	55,629.30	58,890.92	52,260.82	58,568.51
Apr-22	58,530.73	60,845.10	56,009.07	57,060.87
May-22	56,429.45	57,184.21	52,632.48	55,566.41
Jun-22	55,588.27	56,432.65	50,921.22	53,018.94
Jul-22	52,863.34	57,619.27	52,094.25	57,570.25
Aug-22	57,823.10	60,411.20	57,367.47	59,537.07
Sep-22	58,710.53	60,676.12	56,147.23	57,426.92
Oct-22	57,403.92	60,786.70	56,683.40	60,746.59
Nov-22	61,065.58	63,303.01	60,425.47	63,099.65
Dec-22	63,357.99	63,583.07	59,754.10	60,840.74
Jan-23	60,871.24	61,343.96	58,699.20	59,549.90
Feb-23	60,001.17	61,682.25	58,795.97	58,962.12
Mar-23	59,136.48	60,498.48	57,084.91	58,991.52
Apr-23	59,131.16	61,209.46	58,793.08	61,112.44
May-23	61,301.61	63,036.12	61,002.17	62,622.24

Jun-23	62,736.47	64,768.58	62,359.14	64,718.56
Jul-23	64,836.16	67,619.17	64,836.16	66,527.67
Aug-23	66,532.98	66,658.12	64,723.63	64,831.41
Sep-23	64,855.51	67,927.23	64,818.37	65,828.41
Oct-23	65,813.42	66,592.16	63,092.98	63,874.93
Nov-23	63,829.87	67,069.89	63,550.46	66,988.44
Dec-23	67,181.15	72,484.34	67,149.07	72,240.26
Jan-24	72,218.39	73,427.59	70,001.60	71,752.11
Feb-24	71,998.78	73,413.93	70,809.84	72,500.30
Mar-24	72,606.31	74,245.17	71,674.42	73,651.35
Apr-24	73,968.62	75,124.28	73,485.12	74,683.70

YEAR	Date
2010-11	26-Feb-10

Months	Close	Correlation
Jan-10	16,357.96	0.904946523
Feb-10	16,429.55	
Mar-10	17,527.77	
Apr-10	17,558.71	

2011-12	28-Feb-11

Months	Close	Correlation
Jan-11	18,327.76	0.689767841
Feb-11	17,823.40	
Mar-11	19,445.22	
Apr-11	19,135.96	

2012-13	16-Mar-12

Months	Close	Correlation
Jan-12	17,193.55	0.023049566
Feb-12	17,752.68	
Mar-12	17,404.20	
Apr-12	17,318.81	

YEAR	Date
2013-14	28-Feb-13

Months	Close	Correlation
Jan-13	19,894.98	-0.30092
Feb-13	18,861.54	
Mar-13	18,835.77	
Apr-13	19,504.18	

2014-15	10-Jul-14

Months	Close	Correlation
Jun-14	25,413.78	0.945982172
Jul-14	25,894.97	
Aug-14	26,638.11	
Sep-14	26,630.51	

2015-16	28-Feb-15

Months	Close	Correlation
Jan-15	29,182.95	-0.920599623
Feb-15	29,361.50	
Mar-15	27,957.49	
Apr-15	27,011.31	

2016-17	29-Feb-16

Months	Close	Correlation
Jan-16	24,870.69	0.490217287
Feb-16	23,002.00	
Mar-16	25,341.86	
Apr-16	25,606.62	

2017-18	01-Feb-17

Months	Close	Correlation
Jan-17	27,655.96	0.972274901
Feb-17	28,743.32	
Mar-17	29,620.50	
Apr-17	29,918.40	

YEAR	Date
2018-19	01-Feb-18

Months	Close	Correlation
Jan-18	35,965.02	-0.35686
Feb-18	34,184.04	
Mar-18	32,968.68	
Apr-18	35,160.36	

2019-20	05-Jul-19

Months	Close	Correlation
Jun-19	39,394.64	-0.298235875
Jul-19	37,481.12	
Aug-19	37,332.79	
Sep-19	38,667.33	

2020-21	01-Feb-20

Months	Close	Correlation
Jan-20	40,723.49	-0.76566515
Feb-20	38,297.29	
Mar-20	29,468.49	
Apr-20	33,717.62	

2021-22	01-Feb-21

Months	Close	Correlation
Jan-21	46,285.77	0.703880428
Feb-21	49,099.99	
Mar-21	49,509.15	
Apr-21	48,782.36	

2022-23	01-Feb-22

Months	Close	Correlation
Jan-22	58,014.17	-0.087719001
Feb-22	56,247.28	
Mar-22	58,568.51	
Apr-22	57,060.87	

2023-24	01-Feb-23

Months	Close	Correlation
Jan-23	59,549.90	0.607086788
Feb-23	58,962.12	
Mar-23	58,991.52	
Apr-23	61,112.44	

NSE (CNX Nifty) Data and Correlation

Date	Open	High	Low	Close
Jan-10	5200.9	5310.85	4766	4882.05
Feb-10	4882.05	4992	4675.4	4922.3
Mar-10	4935.6	5329.55	4935.35	5249.1
Apr-10	5249.2	5399.65	5160.9	5278
May-10	5278.4	5278.7	4786.45	5086.3
Jun-10	5086.25	5366.75	4961.05	5312.5
Jul-10	5312.05	5477.5	5225.6	5367.6
Aug-10	5369.55	5549.8	5348.9	5402.4
Sep-10	5403.05	6073.5	5403.05	6029.95
Oct-10	6030.3	6284.1	5937.1	6017.7
Nov-10	6092.3	6338.5	5690.35	5862.7
Dec-10	5871	6147.3	5721.15	6134.5
Jan-11	6177.45	6181.05	5416.65	5505.9
Feb-11	5537.3	5599.25	5177.7	5333.25
Mar-11	5382	5872	5348.2	5833.75
Apr-11	5835	5944.45	5693.25	5749.5
May-11	5766.9	5775.25	5328.7	5560.15
Jun-11	5561.05	5657.9	5195.9	5647.4
Jul-11	5705.75	5740.4	5453.95	5482
Aug-11	5527.5	5551.9	4720	5001
Sep-11	5109.8	5169.25	4758.85	4943.25

Oct-11	4874.4	5399.7	4728.3	5326.6
Nov-11	5278.6	5326.45	4639.1	4832.05
Dec-11	4970.85	5099.25	4531.15	4624.3
Jan-12	4640.2	5217	4588.05	5199.25
Feb-12	5198.15	5629.95	5159	5385.2
Mar-12	5366	5499.4	5135.95	5295.55
Apr-12	5296.35	5378.75	5154.3	5248.15
May-12	5254.3	5279.6	4788.95	4924.25
Jun-12	4910.85	5286.25	4770.35	5278.9
Jul-12	5283.85	5348.55	5032.4	5229
Aug-12	5220.7	5448.6	5164.65	5258.5
Sep-12	5276.5	5735.15	5215.7	5703.3
Oct-12	5704.75	5815.35	4888.2	5619.7
Nov-12	5609.85	5885.25	5548.35	5879.85
Dec-12	5878.25	5965.15	5823.15	5905.1
Jan-13	5937.65	6111.8	5935.2	6034.75
Feb-13	6040.95	6052.95	5671.9	5693.05
Mar-13	5702.45	5971.2	5604.85	5682.55
Apr-13	5697.35	5962.3	5477.2	5930.2
May-13	5911.4	6229.45	5910.95	5985.95
Jun-13	5997.35	6011	5566.25	5842.2
Jul-13	5834.1	6093.35	5675.75	5742
Aug-13	5776.9	5808.5	5118.85	5471.8
Sep-13	5480.25	6142.5	5318.9	5735.3
Oct-13	5756.1	6309.05	5700.95	6299.15
Nov-13	6289.75	6342.95	5972.45	6176.1
Dec-13	6171.15	6415.25	6129.95	6304
Jan-14	6323.8	6358.3	6027.25	6089.5
Feb-14	6058.8	6282.7	5933.3	6276.95
Mar-14	6264.35	6730.05	6212.25	6704.2
Apr-14	6729.5	6869.85	6650.4	6696.4
May-14	6709.95	7563.5	6638.55	7229.95
Jun-14	7264.05	7700.05	7239.5	7611.35
Jul-14	7629	7840.95	7422.15	7721.3
Aug-14	7662.5	7968.25	7540.1	7954.35
Sep-14	7990.35	8180.2	7841.8	7964.8
Oct-14	7960.5	8330.75	7723.85	8322.2
Nov-14	8348.15	8617	8290.25	8588.25
Dec-14	8605.1	8626.95	7961.35	8282.7
Jan-15	8272.8	8996.6	8065.45	8808.9
Feb-15	8802.5	8941.1	8470.5	8901.85
Mar-15	8953.85	9119.2	8269.15	8491
Apr-15	8483.7	8844.8	8144.75	8181.5

May-15	8230.05	8489.55	7997.15	8433.65
Jun-15	8417.25	8467.15	7940.3	8368.5
Jul-15	8376.25	8654.75	8315.4	8532.85
Aug-15	8510.65	8621.55	7667.25	7971.3
Sep-15	7907.95	8055	7539.5	7948.9
Oct-15	7992.05	8336.3	7930.65	8065.8
Nov-15	8054.55	8116.1	7714.15	7935.25
Dec-15	7958.15	7979.3	7551.05	7946.35
Jan-16	7938.45	7972.55	7241.5	7563.55
Feb-16	7589.5	7600.45	6825.8	6987.05
Mar-16	7038.25	7777.6	7035.1	7738.4
Apr-16	7718.05	7992	7516.85	7849.8
May-16	7822.7	8213.6	7678.35	8160.1
Jun-16	8179.2	8308.15	7927.05	8287.75
Jul-16	8313.05	8674.7	8287.55	8638.5
Aug-16	8654.3	8819.2	8518.15	8786.2
Sep-16	8793.6	8968.7	8555.2	8611.15
Oct-16	8666.15	8806.95	8506.15	8625.7
Nov-16	8653.15	8669.6	7916.4	8224.5
Dec-16	8244	8274.95	7893.8	8185.8
Jan-17	8210.1	8672.7	8133.8	8561.3
Feb-17	8570.35	8982.15	8537.5	8879.6
Mar-17	8904.4	9218.4	8860.1	9173.75
Apr-17	9220.6	9367.15	9075.15	9304.05
May-17	9339.85	9649.6	9269.9	9621.25
Jun-17	9603.55	9709.3	9448.75	9520.9
Jul-17	9587.95	10114.85	9543.55	10077.1
Aug-17	10101.05	10137.85	9685.55	9917.9
Sep-17	9937.65	10178.95	9687.55	9788.6
Oct-17	9893.3	10384.5	9831.05	10335.3
Nov-17	10390.35	10490.45	10094	10226.55
Dec-17	10263.7	10552.4	10033.35	10530.7
Jan-18	10531.7	11171.55	10404.65	11027.7
Feb-18	11044.55	11117.35	10276.3	10492.85
Mar-18	10479.95	10525.5	9951.9	10113.7
Apr-18	10151.65	10759	10111.3	10739.35
May-18	10783.85	10929.2	10417.8	10736.15
Jun-18	10738.45	10893.25	10550.9	10714.3
Jul-18	10732.35	11366	10604.65	11356.5
Aug-18	11359.8	11760.2	11234.95	11680.5
Sep-18	11751.8	11751.8	10850.3	10930.45
Oct-18	10930.9	11035.65	10004.55	10386.6
Nov-18	10441.7	10922.45	10341.9	10876.75

Dec-18	10930.7	10985.15	10333.85	10862.55
Jan-19	10881.7	10987.45	10583.65	10830.95
Feb-19	10851.35	11118.1	10585.65	10792.5
Mar-19	10842.65	11630.35	10817	11623.9
Apr-19	11665.2	11856.15	11549.1	11748.15
May-19	11725.55	12041.15	11108.3	11922.8
Jun-19	11953.75	12103.05	11625.1	11788.85
Jul-19	11839.9	11981.75	10999.4	11118
Aug-19	11060.2	11181.45	10637.15	11023.25
Sep-19	10960.95	11694.85	10670.25	11474.45
Oct-19	11515.4	11945	11090.15	11877.45
Nov-19	11886.6	12158.8	11802.65	12056.05
Dec-19	12137.05	12293.9	11832.3	12168.45
Jan-20	12202.15	12430.5	11929.6	11962.1
Feb-20	11939	12246.7	11175.05	11201.75
Mar-20	11387.35	11433	7511.1	8597.75
Apr-20	8584.1	9889.05	8055.8	9859.9
May-20	9533.5	9598.85	8806.75	9580.3
Jun-20	9726.85	10553.15	9544.35	10302.1
Jul-20	10323.8	11341.4	10299.6	11073.45
Aug-20	11057.55	11794.25	10882.25	11387.5
Sep-20	11464.3	11618.1	10790.2	11247.55
Oct-20	11364.45	12025.45	11347.05	11642.4
Nov-20	11697.35	13145.85	11557.4	12968.95
Dec-20	13062.2	14024.85	12962.8	13981.75
Jan-21	13996.1	14753.55	13596.75	13634.6
Feb-21	13758.6	15431.75	13661.75	14529.15
Mar-21	14702.5	15336.3	14264.4	14690.7
Apr-21	14798.4	15044.35	14151.4	14631.1
May-21	14481.05	15606.35	14416.25	15582.8
Jun-21	15629.65	15915.65	15450.9	15721.5
Jul-21	15755.05	15962.25	15513.45	15763.05
Aug-21	15874.9	17153.5	15834.65	17132.2
Sep-21	17185.6	17947.65	17055.05	17618.15
Oct-21	17531.9	18604.45	17452.9	17671.65
Nov-21	17783.15	18210.15	16782.4	16983.2
Dec-21	17104.4	17639.5	16410.2	17354.05
Jan-22	17387.15	18350.95	16836.8	17339.85
Feb-22	17529.45	17794.6	16203.25	16793.9
Mar-22	16593.1	17559.8	15671.45	17464.75
Apr-22	17436.9	18114.65	16824.7	17102.55
May-22	16924.45	17132.85	15735.75	16584.55
Jun-22	16594.4	16793.85	15183.4	15780.25

Jul-22	15703.7	17172.8	15511.05	17158.25
Aug-22	17243.2	17992.2	17154.8	17759.3
Sep-22	17485.7	18096.15	16747.7	17094.35
Oct-22	17102.1	18022.8	16855.55	18012.2
Nov-22	18130.7	18816.05	17959.2	18758.35
Dec-22	18871.95	18887.6	17774.25	18105.3
Jan-23	18131.7	18251.95	17405.55	17662.15
Feb-23	17811.6	18134.75	17255.2	17303.95
Mar-23	17360.1	17799.95	16828.35	17359.75
Apr-23	17427.95	18089.15	17312.75	18065
May-23	18124.8	18662.45	18042.4	18534.4
Jun-23	18579.4	19201.7	18464.55	19189.05
Jul-23	19246.5	19991.85	19234.4	19753.8
Aug-23	19784	19795.6	19223.65	19253.8
Sep-23	19258.15	20222.45	19255.7	19638.3
Oct-23	19622.4	19849.75	18837.85	19079.6
Nov-23	19064.05	20158.7	18973.7	20133.15
Dec-23	20194.1	21801.45	20183.7	21731.4
Jan-24	21727.75	22124.15	21137.2	21725.7
Feb-24	21780.65	22297.5	21530.2	21982.8
Mar-24	22048.3	22526.6	21710.2	22326.9
Apr-24	22455	22768.4	22303.8	22642.75

YEAR	Date
2010 - 11	26-Feb-10

Months	Close	Correlation
Jan-10	5310.85	0.416466581
Feb-10	4992	
Mar-10	5329.55	
Apr-10	5399.65	

2011 - 12	28-Feb-11

Months	Close	Correlation
Jan-11	6181.05	-0.24690135
Feb-11	5599.25	
Mar-11	5872	
Apr-11	5944.45	

2012 - 13	16-Mar-12

Months	Close	Correlation
Feb-12	5629.95	-0.997737724
Mar-12	5499.4	
Apr-12	5378.75	
May-12	5279.6	

2013 - 14	28-Feb-13

Months	Close	Correlation
Jan-13	6111.8	-0.959902306
Feb-13	6052.95	
Mar-13	5971.2	
Apr-13	5962.3	

YEAR	Date
2014 - 15	10-Jul-14

Months	Close	Correlation	
Jun-14	7700.05	0.993474874	
Jul-14	7840.95		
Aug-14	7968.25		
Sep-14	8180.2		

2015 - 16	28-Feb-15

Months	Close	Correlation
Jan-15	8996.6	-0.328041406
Feb-15	8941.1	
Mar-15	9119.2	
Apr-15	8844.8	

2016 - 17	29-Feb-16

Months	Close	Correlation
Jan-16	7972.55	0.16021856
Feb-16	7600.45	
Mar-16	7777.6	
Apr-16	7992	

2017 - 18	01-Feb-17

Months	Close	Correlation
Jan-17	8672.7	0.987871508
Feb-17	8982.15	
Mar-17	9218.4	
Apr-17	9367.15	

2018 - 19	01-Feb-18

Months	Close	Correlation
Jan-18	11171.55	-0.760086061
Feb-18	11117.35	
Mar-18	10525.5	
Apr-18	10759	

YEAR	Date
2019 - 20	05-Jul-19

Months	Close	Correlation
Jun-19	12103.05	-0.636384749
Jul-19	11981.75	
Aug-19	11181.45	
Sep-19	11694.85	

2020 - 21	01-Feb-20

Months	Close	Correlation
Jan-20	12430.5	-0.940763023

Feb-20	12246.7
Mar-20	11433
Apr-20	9889.05

2021 - 22	01-Feb-21

Months	Close	Correlation
Jan-21	14753.55	0.331901098
Feb-21	15431.75	
Mar-21	15336.3	
Apr-21	15044.35	

2022 - 23	01-Feb-22

Months	Close	Correlation
Jan-22	18350.95	-0.346111716
Feb-22	17794.6	
Mar-22	17559.8	
Apr-22	18114.65	

2023 - 24	01-Feb-23

Months	Close	Correlation
Jan-23	18251.95	-0.54202093
Feb-23	18134.75	
Mar-23	17799.95	
Apr-23	18089.15	